

QuantERA Proposal Template

Project Acronym

PA-QML

Project Title

Physics-Aware Quantum Machine Learning: From Distributed Hardware to Certified Intelligence

Topic:



Quantum Phenomena and Resources



Applied Quantum Science

Coordinator contact point for the proposal

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People involved in the realisation of the project (please enter these names and their email addresses in the financial template as well)

Partner Number	Country	Institution/Department	Name of the Principal Investigator (PI) ¹	Name of the co-Investigators ²	Name of the other personnel participating in the project ³
1 Coordinator	South Korea	Seoul National University / Department of Psychology; Department of Brain and Cognitive Sciences; Interdisciplinary Program in Artificial Intelligence	Jiook Cha		Junghoon Justin Park; Maria Pak; Seung Yun Choi; Eunji Lee; Bo-Gyeom Kim; Heehwan Wang; Sangyoon Bae; Jungwoo Seo; Seungju Lee; Jubin Choi; Danny

¹ The Principal Investigator (PI) is the point of contact of the partner for the corresponding Research Funding Organisation.

² A co-investigator is a known scientist and/or group leader making a substantial contribution to the project.

³ If the name is for the moment unknown, specify the level of expertise sought (PhD, post-doc, engineer, professor...).

					Dongyeop Han; Kyungjin Oh; Ahhyun Lucy Lee; Seok Jin Moon; Sebin Lee; Yujin Yun; Allison Eun Se You; Bohee Lee
2	South Korea	Yonsei University / Department of Physics	Hwidong Yoo		Sungwon Kim; Yun Eo; Haeun Jang
3	Germany	Fraunhofer / Fraunhofer Institute for Cognitive Systems IKS	Jeanette Miriam Lorenz		Alona Sakhnenko
4	Italy	University of Naples Federico II	Giovanni Acampora	Roberto Schiattarella	Autilia Vitiello; Angela Chiatto

Duration: 36 months

Summary of the project (publishable abstract, max. 1/2 page):

Be precise and concise. This summary will be used to select suited reviewers for the proposal.

Relevance to the topic addressed in the call (in particular specify here which part of the call text is addressed by your project, max. 1/4 page):

Be precise and concise.

1. Excellence (max. 6 pages)

1.1 Targeted Breakthrough, Baseline of Knowledge and Skills

Targeted Breakthroughs

This project targets **four fundamental barriers** preventing Quantum Machine Learning (QML) from achieving practical advantage. Current QML remains confined to small-scale demonstrations reliant on heuristic ansatzes that become untrainable at scale. We shift the paradigm from "heuristic QML" to "**Physics-Aware & Certified QML**"—algorithms designed for NISQ constraints rather than ideal hardware assumptions.

Breakthrough 1 – Scalability via Multi-Chip Ensembles: We partition high-dimensional data across k independent QPUs, each processing locally-correlated inputs. **Selective Entanglement** introduces inter-chip quantum connections only for globally-dependent features. Classical aggregation provides algorithmic error mitigation: independent chip errors average out (variance reduction $\propto 1/k$) without resource-heavy correction codes, establishing a **resource theory for "Collective Quantum Advantage"**. This architecture aligns with emerging modular quantum hardware platforms (e.g., IBM Quantum, IonQ, Pasqal), enabling direct deployment as interconnected QPU systems become available.

Breakthrough 2 – Trainability via QFF-HQGA: We integrate **Quantum Forward-Forward (QFF)** with **Hybrid Quantum Genetic Algorithm (HQGA)**. QFF decomposes deep circuits into layers optimizing local goodness objectives, mathematically circumventing barren plateaus while preserving expressivity. HQGA encodes parameters as **quantum chromosomes**—superpositions enabling parallel fitness evaluation—using **entangled crossover** and **quantum elitism**.

Breakthrough 3 – Temporal Expressibility via Q-SSM: We develop **Quantum State Space Models (Q-SSM)** integrating three-branch quantum superposition with LSTM-style gating. Quantum circuits access 2^n -dimensional Hilbert space with $O(n \times l)$ parameters versus $O(N^2)$ classically. The architecture achieves **$O(L)$ complexity** through chunked processing, matching classical Mamba/Hydra efficiency while exploiting quantum feature expressivity.

Breakthrough 4 – Reliability via Fuzzy Quantum Diffusion: We exploit physical quantum noise as a generative resource. **Fuzzy Quantum Logic** via POVMs provides continuous measurements, enabling **Hardware-Attention U-Net** to learn device-specific noise fingerprints. To curb uncertainty from noisy models further, we design **robustness and safety certification** that meet **industrial-grade robustness** standards, with verified resilience to targeted attacks and untargeted perturbations.

Consortium Expertise:

Partner	Expertise	Project Role
SNU (Cha)	Multi-Chip Ensembles, Q-SSM, Quantum Transformers, HQTCN	Coordinator; Architecture Lead (WP1, WP2)
Naples (Acampora)	Evolutionary Algorithms, Fuzzy Logic, Quantum Computational Intelligence	Optimization & Genetic Model Lead (WP3)
Fraunhofer IKS (Lorenz)	QML, robustness certification, European QC Benchmarking Committee	Certification Lead (WP5, WP6)

Partner	Expertise	Project Role
Yonsei (Yoo)	CMS/CERN collaboration, Q-ABCDISCo, TCN-VQC, Quantum ViT for HEP	Domain Validation Lead (WP4)

Specific Objectives

Objective	Goal	Measure	Target	Timeline
O1: Scalability	Multi-Chip Ensembles with Selective Entanglement	Accuracy retention vs monolithic circuit	$\geq 90\%$ accuracy at $2\times$ effective qubit scaling; $< 10\%$ variance across configurations	M12→M30
O2: Trainability	QFF-HQGA hybrid optimization	Convergence in barren plateau regimes; measurement count	Convergence at > 10 layers where BP fails; $\geq 50\%$ fewer circuit evaluations vs parameter-shift	M12→M24
O3: Temporal	Q-SSM with three-branch superposition	Memory capacity; complexity scaling	$\geq 1.5\times$ memory capacity vs classical SSM; verified $O(L)$ scaling	M18→M30
O4: Reliability	Fuzzy Quantum Diffusion + safety certification	FID under noise; Lipschitz constant	$FID \leq 1.5\times$ classical baseline under 10^{-3} gate error; Lipschitz $L < 2.0$	M18→M36
O5: Validation	Domain science certification of O1-O4	HEP/Neuro/Cyber benchmarks	Statistical quantum advantage ($p < 0.05$) in ≥ 1 domain metric	M6→M36

O5 Validation Details:

- **HEP:** Multi-Chip Quantum ViT for calorimeter classification; Q-ABCDISCo ($\sim 1,372$ parameters) with Distance Correlation penalty for background estimation; TCN-VQC (4 qubits) with Noise2Noise self-supervision for waveform denoising
- **Neuroscience:** Q-SSM and Quantum Transformers for EEG/fMRI spatio-temporal analysis (→O3); QFF-HQGA optimized QAOA for brain network modularity optimization and VQE for active inference modeling (→O2)
- **Cybersecurity:** Fuzzy Quantum Diffusion for anomaly detection via hardware fingerprinting; certified robustness against targeted (e.g. adversarial attacks) and untargeted (e.g. input or device noise) perturbations.

1.2 Novelty, Level of Ambition and Foundational Character

Advance Beyond State-of-the-Art

Challenge	SOTA Limitation	Our Advance
Scalability	Circuit cutting incurs exponential sampling	Resource theory for Collective Quantum Advantage: ensemble aggregation preserves expressivity; partitioning eliminates inter-chip

Challenge	SOTA Limitation	Our Advance
	overhead; no multi-modal fusion theory exists	entanglement costs; Selective Entanglement optimizes expressivity-trainability trade-off
Trainability	Gradient-based: barren plateaus at $O(n)$ depth. Gradient-free: slow convergence, high sampling. Trainable circuits often classically simulable	Symbiotic QFF-HQGA: local goodness objectives mathematically avoid BP; quantum chromosomes with entangled crossover navigate non-simulable deep circuit landscapes without gradient computation
Temporal	Quantum RNNs collapse sequences via pooling; no quantum SSM exists	Novel Q-SSM architecture: measurement-based superposition ($\alpha, \beta, \gamma \in \mathbb{C}$) preserves gradient flow; LSTM gating enables selective memory; chunked processing achieves $O(L)$ complexity matching classical Mamba/Hydra
Reliability	Noise treated as error requiring resource-heavy correction; no QML certification standards	Noise-as-resource paradigm for QML: Fuzzy POVMs enable hardware-native generative modeling; safety certification protocol with formal guarantees.

Level of Ambition

Foundational, not incremental: We establish new theoretical frameworks—resource theory for distributed QML, mathematical proof of barren plateau avoidance via local objectives, formal certification metrics—rather than algorithmic refinements.

"Big Science" Validation: We validate on datasets with genuine complexity, not toy benchmarks:

- **HEP (LHC/CMS):** Petabyte-scale data with complex non-local correlations. Q-ABCDiCo achieves background estimation with $\sim 1,372$ parameters (vs $\sim 10^5$ classical). Quantum ViT targets merged electron classification at $\Delta R \approx 0.001$ resolution.
- **Neuroscience:** High-dimensional, data-scarce clinical signals. Q-SSM captures long-range temporal patterns in EEG/fMRI. QFF-HQGA enables training of QAOA for brain network modularity optimization and VQE for active inference modelling.
- **Cybersecurity:** Dynamic threat landscapes requiring certified robustness. QUARK provides formal adversarial guarantees unprecedented in QML.

Industrial Certification Standard: we transform QML from experimental demonstrations to certifiable engineering—a prerequisite for safety-critical deployment in automotive, medical, and financial sectors.

1.3 Concept and Methodology

Overall Concept: Physics-Aware QML Stack

We assume NISQ devices remain fragmented and noisy. Rather than waiting for fault tolerance, we redesign algorithms to exploit these constraints.

Core hypothesis: Quantum advantage lies in *feature representation* (Hilbert space expressivity with parameter efficiency), not error-free computation. Our hybrid architectures extract quantum features via shallow, noise-resilient circuits while delegating temporal memory and output generation to classical components proven effective for these tasks.

Methodology

M1: Multi-Chip Ensembles (→O1: Scalability)

Mechanism: Partition input $x \in \mathbb{R}^d$ into k subsets $\{x_1, \dots, x_k\}$ assigned to independent n -qubit chips. Each chip i applies modality-specific ansatz $U_i(\theta_i)$. **Selective Entanglement**

introduces controlled inter-chip operations only between features with high classical correlation (identified via mutual information analysis). Outputs aggregate via learned weights $\hat{y} = \sum_i w_i \cdot f(\langle \psi_i | M | \psi_i \rangle)$.

Scientific basis: Partitioning restricts entanglement entropy growth across the full system, provably mitigating barren plateaus (concentration of measure scales with entangled subsystem size, not total qubits). Independent chip errors are uncorrelated; aggregation reduces variance by factor $1/k$ without correction overhead.

M2: QFF-HQGA Optimization (→O2: Trainability)

Mechanism: Decompose L -layer circuit into L independent optimization problems. Layer l minimizes local goodness $G_l(\theta_l) = \langle \psi_{pos} | M_l | \psi_{pos} \rangle - \langle \psi_{neg} | M_l | \psi_{neg} \rangle$, where ψ_{pos}/ψ_{neg} are states from positive (real) and negative (corrupted) data passes. Within each layer, HQGA encodes parameters as quantum chromosomes $|\theta\rangle = \sum_j \alpha_j |\theta_j\rangle$. **Entangled crossover**

(controlled rotations correlating successful parameter traits) and **quantum elitism** (amplitude amplification of high-fitness states) evolve the population.

Scientific basis: Local objectives have $O(1)$ gradient variance—versus exponentially vanishing global gradients—because each layer's loss landscape is shallow. Quantum parallelism evaluates population fitness in $O(1)$ circuit executions per generation, eliminating the measurement overhead of parameter-shift gradient estimation.

M3: Q-SSM Architecture (→O3: Temporal Expressibility)

Mechanism: For sequence $\{x_t\}$, partition into chunks. Each chunk passes through three parallel VQCs producing measurements $\{m_\alpha, m_\beta, m_\gamma\}$. Combine via trainable complex coefficients:

$$h_q = \alpha \cdot m_\alpha + \beta \cdot m_\beta + \gamma \cdot m_\gamma, \text{ where } \alpha, \beta, \gamma \in \mathbb{C}$$

This **measurement-based superposition** enables interference-like feature combination while maintaining gradient flow. Classical LSTM gates process temporal dynamics:

$$c_t = f_t \odot c_{t-1} + i_t \odot h_q o_t = \tanh(c_t) \odot \sigma(W_o \cdot h_q)$$

Chunked processing divides long sequences into manageable segments, with LSTM gates maintaining state across chunks—achieving $O(L)$ complexity analogous to classical Mamba/Hydra while leveraging quantum Hilbert space expressivity.

Scientific basis: Quantum circuits provide 2^n -dimensional feature space with $O(n \times l)$ parameters versus $O(N^2)$ for equivalent classical layers. LSTM gating—proven optimal for temporal credit assignment—handles sequence memory. Complexity is $O(L)$ in sequence length (constant quantum cost per chunk).

M4: Fuzzy Quantum Diffusion + Reliability & Safety Certification (→O4: Reliability)

Mechanism: Forward diffusion uses physical noise channels (amplitude damping, dephasing, crosstalk) instead of synthetic Gaussian noise. **POVMs** provide continuous "fuzzy" measurements $x_t = \text{Tr}[E_t(\rho) \cdot E_{\text{detect}}]$ where E_t parameterizes measurement sharpness by decoherence time t . **Hardware-Attention U-Net** learns device-specific reversal using calibration maps (T_1, T_2 , readout error rates). **Certification** computes:

- **Lipschitz constant:** $L = \max_{\{x, x'\}} \frac{\|f(x) - f(x')\|}{\|x - x'\|}$ (certified robustness bound)
- **Gradient variance** under noise (trainability metric)
- **Entangling capability** (expressivity metric)

Scientific basis: Physical noise is deterministic given device characterization—unlike stochastic Gaussian—enabling precise reversal learning. Fuzzy measurements preserve information lost in sharp projection, maintaining gradient flow for hybrid training. Lipschitz bounds provide formal guarantees: $\text{output perturbation} \leq L \times \text{input perturbation}$.

M5: Domain Validation (→O5)

Domain	Task	Method	Quantum Advantage Hypothesis
HEP	Background estimation	Q-ABCDisCo + DisCo penalty	Parameter efficiency (~1,372 vs ~10 ⁵)
	Waveform denoising	TCN-VQC (4 qubits) + Noise2Noise	Self-supervised learning without clean labels
	Merged electron ID	Multi-Chip Quantum ViT	$\Delta R \approx 0.001$ resolution via quantum attention
Neuro	EEG/fMRI analysis	Q-SSM + Quantum Transformers	Temporal expressivity (→O3)
	Brain network modularity optimization	QFF-HQGA optimized QAOA/VQE	Trainability for combinatorial optimization (→O2)
Cyber	Anomaly detection	Fuzzy Quantum Diffusion	Hardware fingerprint sensitivity
	Robustness certification	Adversarial testing	Formal Lipschitz guarantees

Risk Mitigation

Risk	Likelihood	Mitigation Strategy
Barren plateaus persist despite QFF	Medium	Fallback to pure HQGA (entirely gradient-free); reduce circuit depth
Ensemble accuracy degradation	Medium	Weighted voting by per-chip reliability scores; adaptive chip selection
Hardware noise exceeds Fuzzy model capacity	Low	Pivot to Fuzzy Quantum GANs with stabilized training dynamics
Domain data access delays	Low	CERN Open Data Portal; PhysioNet EEG; public intrusion detection datasets

1.4 Interdisciplinary Nature

Research Disciplines and Integration

Discipline	Contribution	Integration Point
Quantum Information Science	Ansatz design, entanglement theory, variational circuits	Provides computational substrate and expressivity theory

Discipline	Contribution	Integration Point
Computational Intelligence	Fuzzy Logic, Evolutionary Algorithms	Solves trainability/robustness where pure QIS methods fail
Software Engineering & Reliability	benchmarking, certification protocols	Transforms theory into deployable, verifiable standards
Domain Sciences (HEP, Neuroscience)	Complex real-world datasets and domain constraints	Provides validation complexity; grounds quantum advantage claims

Cross-Fertilization: Annual methodology exchanges ensure knowledge transfer (Naples: Fuzzy/evolutionary methods; SNU: Q-SSM/Multi-Chip protocols; Fraunhofer: Safety certification). Domain Challenge Sprints apply theoretical advances to real problems—HEP jet classification, EEG/fMRI temporal modelling, brain network modularity optimization, adversarial robustness testing—evaluated by domain-specific metrics.

Interdisciplinary Synergies:

- **Fuzzy Logic + Quantum Mechanics:** Biological uncertainty (neuroscience) and hardware noise (NISQ) unified as continuous "degrees of truth" via POVMs
- **Evolutionary Algorithms + Quantum Circuits:** QML models that *evolve* through non-convex landscapes where gradient-based optimization fails
- **Distributed Computing + HEP:** Multi-Chip protocols validated on particle data with intrinsic non-local correlations—ideal for demonstrating ensemble aggregation preserves quantum expressivity

A unified software architecture with standardized interfaces (Physics, Logic, Certification modules) ensures integration at the code level.

1.5 Gender Dimension and Open Science Practices

Gender Dimension

The foundational QML algorithms are gender-neutral. For the **Neuroscience validation (O5)**, where EEG/fMRI data involves human subjects, we implement **gender-stratified validation**: balanced training data (50/50 male/female) and sex as an explicit covariate to prevent algorithmic bias in brain signal analysis.

The consortium includes **female leadership** (PD Dr. habil. Jeanette Miriam Lorenz, Fraunhofer IKS) as Principal Investigator, serving as a role model for early-career researchers.

Open Science Practices

- **Open Access:** 100% of publications via Green (arXiv, Zenodo) or Gold routes.
- **FAIR Data:** Challenge datasets (anonymized EEG, LHC subsets) deposited in Zenodo with DOIs. Data Management Plan delivered by M3.
- **Open Source:** "Physics-Aware QML" library released under Apache 2.0 license on GitHub, with Jupyter notebook tutorials for reproducibility.
- **Sensitive Data:** Neuroscience data pseudonymized at source (GDPR/PIPA compliant). Cybersecurity logs screened for PII before consortium sharing.

2. Impact (max. 3 pages)

2.1 Expected Impacts

This project addresses the QuantERA QPR call objective to *"explore novel quantum phenomena and resources... and address major challenges preventing broad applications."* We establish a **resource theory for scalable and reliable QML** with the following impacts:

Quantified Impact Indicators

Impact	Objective	Indicator	Baseline	Target	Timeline
Distributed Computing	O1	Accuracy retention at 2× qubit scaling	Single-chip: ~20-50 qubits	$\geq 90\%$, variance $\propto 1/k$	M30
Trainability	O2	Convergence in barren plateau regimes	Gradient methods fail >10 layers	QFF-HQGA converges >10 layers	M24
Temporal Learning	O3	Memory capacity vs. classical SSM	Quantum RNNs collapse sequences	$\geq 1.5\times$ classical; $O(L)$ complexity	M30
Reliability	O4	Certified robustness (Lipschitz bound)	No QML certification exists	$L < 2.0$; FID $\leq 1.5\times$ at 10^{-3} error	M36

Key Contributions:

- **O1:** "Collective Quantum Advantage" via Multi-Chip Ensembles [Park et al., 2025]—ensemble aggregation provides algorithmic error mitigation (variance $\propto 1/k$) without correction codes
- **O2:** QFF-HQGA resolves the optimization trilemma—local goodness objectives [Hinton, 2022] avoid barren plateaus; quantum evolutionary search [Acampora & Vitiello, 2021] eliminates gradient overhead
- **O3:** Q-SSM [Park et al., 2025b] achieves $O(L)$ complexity matching classical Mamba [Gu & Dao, 2023] while exploiting 2^n -dimensional Hilbert space with $O(n \times l)$ parameters
- **O4:** Safety certification [Kiwit et al., 2024; Lorenz et al., 2025] provides formal Lipschitz guarantees; Fuzzy POVMs [Nunziata et al., 2025] enable noise-as-resource paradigm [Parigi et al., 2024]

Domain Science Validation (→O5)

Domain	Task	Method	Quantum Advantage Hypothesis	Success Metric
HEP	Background estimation	Q-ABCDIsCo	$\sim 1,372$ vs. $\sim 10^5$ classical parameters	$p < 0.05$
	Waveform denoising	TCN-VQC + Noise2Noise	Self-supervised without clean labels	SNR improvement
	Merged electron ID	Multi-Chip Quantum ViT	$\Delta R \approx 0.001$ resolution	Classification accuracy

Domain	Task	Method	Quantum Advantage Hypothesis	Success Metric
Neuro	EEG/fMRI analysis	Q-SSM	Long-range temporal patterns	Memory $\geq 1.5\times$ classical
	Brain network modularity optimization	QFF-HQGA QAOA/VQE	Trainability for combinatorial problems [Friston et al., 2017]	Convergence
Cyber	Anomaly detection	Fuzzy Quantum Diffusion	Hardware fingerprint sensitivity	Detection rate
	Robustness certification	Noise in QML [Winderl et al., 2024]	Formal adversarial guarantees	$L < 2.0$

Transformational Societal Impact

Impact Area	Gap Addressed	Our Contribution	Societal Benefit
Industrial Certification	No QML safety/robustness standards	Pipeline-integrated certification (Lipschitz bounds, noise stability)	Enables quantum AI in safety-critical sectors (automotive, medical, finance)
Democratization	Quantum advantage restricted to elite labs	Multi-Chip Ensembles aggregate smaller QPUs	Universities/SMEs access "virtual large-scale" QML
Privacy-Preserving Healthcare	Medical data sharing constrained by GDPR/HIPAA	Fuzzy Quantum Diffusion generates synthetic data [Zhang et al., 2024]	Unlocks medical AI development without privacy compromise
Fundamental Physics	LHC analysis bottlenecks	Quantum ViT/Q-ABCDiSCo with superior parameter efficiency	Accelerates discovery beyond Standard Model

2.2 Dissemination, Exploitation, and Communication

Dissemination Strategy

- Publications (≥ 12 peer-reviewed, 100% Open Access):

Type	Venues	Timeline	Lead
Theory	<i>Nature Comms, PRX Quantum</i>	M12-M24	SNU, All
Methods	<i>npj Quantum Info, QST</i>	M18-M30	SNU, Naples
Applications	<i>PRL, NeuroImage, IEEE S&P</i>	M24-M36	Yonsei, Fraunhofer

Type	Venues	Timeline	Lead
Tools	<i>Quantum Machine Intelligence</i>	M30-M36	Fraunhofer, Naples

- **Conferences:** IEEE QAI, IEEE QCE (2-3 papers each); NeurIPS/ICML workshops; domain conferences (CHEP, OHBM, IEEE S&P)
- **Training:** Annual consortium workshops (M12, M24, M36); Joint summer school "Practical QML for Domain Scientists" (M18)

Exploitation Plan

Innovation	IP Strategy	Pathway
Multi-Chip Ensemble Protocol	Open-source (Apache 2.0)	PennyLane/Qiskit integration
QFF-HQGA Optimizer	Open-source	Community adoption
Certification Extensions to Software Pipelines	Joint IP (Fraunhofer)	Industrial licensing
Domain Models & Datasets	Open (Zenodo, HuggingFace)	Benchmark standardization

- **Software Deliverables:** QML-Ensemble Library (GitHub); Certification Module; Domain Benchmark Suites (Q-ABCDiCo, Q-SSM-EEG, Fuzzy-QIDS)
- **Commercialization:** Hardware vendors (IBM, IQM, Pasqal)—Multi-Chip protocols; Pharma/Biotech—synthetic data licensing; Automotive/Aerospace—Fraunhofer certification consulting
- **Standardization:** IEEE P7131 (Quantum Performance Metrics); QED-C benchmarking; ISO/IEC JTC 1/SC 42 (AI certification)

Communication Activities

Audience	Channels	KPIs
Scientific	Publications, GitHub, conferences	500+ GitHub stars, 100+ forks
Industry	Workshops, white papers, advisory board	200+ webinar registrations
Policymakers	Policy briefs (M18, M36)	50+ decision-makers reached
Public	Website, social media, video series	5,000+ visitors, 10,000+ views

Data Management (FAIR Principles)

Data Type	Repository	Access	Timeline
Benchmark datasets	Zenodo (DOI)	Open	M24
Trained models	HuggingFace	Open	M30
Source code	GitHub	Apache 2.0	Continuous
Raw data	Institutional	Restricted	M36

Reproducibility: Docker containers, Jupyter notebooks, OSF pre-registration for validation experiments.

3. Implementation (Weight: 25%)

3.1 Work Plan (max. 2 pages)

Overall Structure

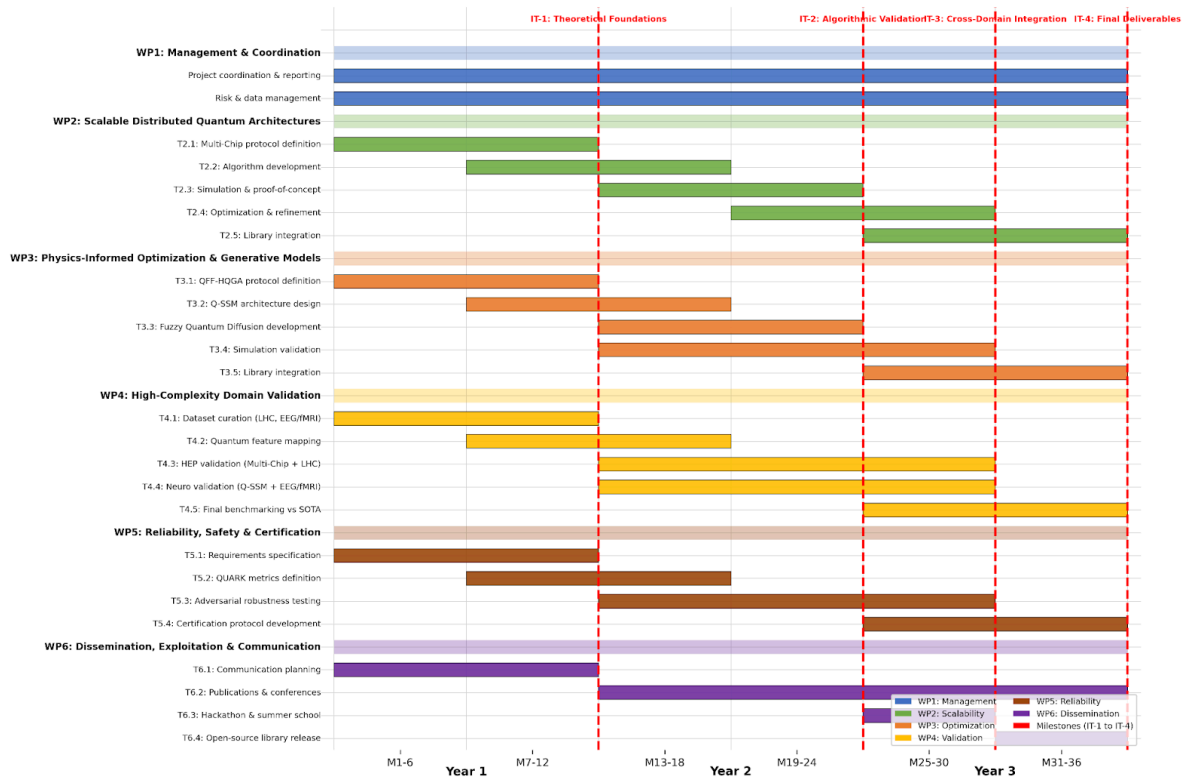
The project is organized into **six interconnected Work Packages (WPs)** creating a feedback loop between foundational theory development and empirical validation:

WP	Title	Lead	Focus
WP1	Management & Coordination	SNU	Scientific alignment, reporting, risk/data management
WP2	Scalable Distributed Quantum Architectures	SNU	Multi-Chip Ensemble protocols (→O1)
WP3	Physics-Informed Optimization & Generative Models	Naples	QFF-HQGA, Q-SSM, and Fuzzy Quantum Diffusion (→O2, O3, O4)
WP4	High-Complexity Domain Validation	Yonsei	HEP and Neuroscience benchmarking (→O5)
WP5	Reliability, Safety & Certification	Fraunhofer	Robustness certification (→O4)
WP6	Dissemination, Exploitation & Communication	Fraunhofer	Open Science, software release, outreach

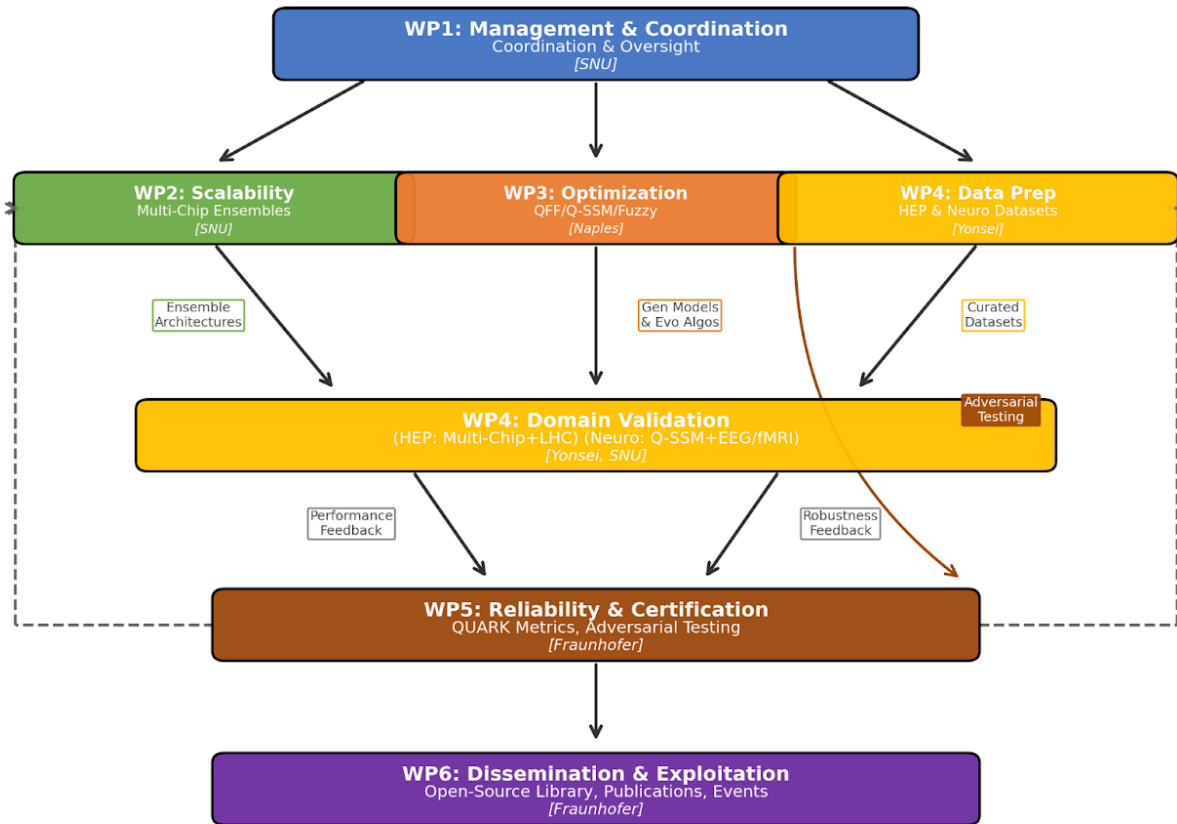
Intermediate Targets

Target	Month	Description	Success Criteria
IT-1	M12	Theoretical Foundations Established	Multi-Chip protocol specs complete; Q-SSM architecture defined; Fuzzy Logic Controllers specified; Domain datasets preprocessed
IT-2	M24	Algorithmic Validation (Simulation)	QFF-HQGA converges on >10-layer circuits where backpropagation fails; Q-SSM achieves $\geq 1.5\times$ classical memory capacity; Fuzzy-Diffusion achieves FID $\leq 1.5\times$ classical diffusion baseline
IT-3	M30	Cross-Domain Integration	Multi-Chip Ensemble reconstructs LHC particle correlations; Cybersecurity models pass QUARK robustness certification
IT-4	M36	Final Deliverables	Open-source "Physics-Aware QML" library released; Final benchmarks published

Gantt Chart: Work Package Timeline



PERT Diagram: Work Package Dependencies



Validation feedback refines foundational algorithms (WP4/WP5 → WP2/WP3)

Key Dependencies:

- **WP2 → WP4:** Multi-Chip Ensemble protocols feed into HEP scalability validation
- **WP3 → WP4:** Q-SSM feeds into Neuro spatio-temporal analysis; QFF-HQGA feeds into brain network modularity optimization; Fuzzy-Diffusion feeds into Cyber anomaly detection
- **WP3 → WP5:** Evolutionary QML models undergo adversarial robustness testing
- **WP4/WP5 → WP2/WP3:** Validation feedback refines foundational algorithms
- **All → WP6:** Validated results flow to dissemination

3.2 Work Packages

WP 1	Project Management & Coordination				Start Month: 1	End Month: 36
Contribution of project partners						
Partner number ⁴		1. SNU	2. Yonsei	3. Fraunhofer	4. Naples	
Total effort per partner (person*months)		18	2	4	2	
Aim of the WP						
To ensure efficient administrative and scientific coordination, timely reporting to national funding agencies and the QuantERA secretariat, and effective risk/data management across the consortium.						
Tasks						
T1.1	Consortium Coordination & Monitoring (M1–M36; SNU; All) ⁵ Day-to-day management of the project; organization of bi-annual consortium meetings (virtual/physical); monitoring of milestones and KPIs; preparation of periodic progress reports.					
T1.2	Risk & Innovation Management (M1–M36; SNU; Fraunhofer) Continuous monitoring of scientific and technical risks; execution of mitigation plans; management of Intellectual Property Rights (IPR); coordination of innovation activities.					
T1.3	Data Management & Ethics (M1–M36; SNU; Fraunhofer) Implementation of the Data Management Plan (DMP) ensuring FAIR principles; oversight of ethical compliance regarding EEG/fMRI data (Neuroscience) and sensitive network logs (Cybersecurity); GDPR compliance monitoring.					
Deliverable		Month of delivery	Title of deliverable			
D1.1		M3	Data Management Plan			
D1.2		M18	Mid-term Progress Report			
D1.3		M36	Final Project Report			

⁴ **Bold** the partner number of the work package leader

⁵ For instance: T1.1 Development of something (M3-M6; responsible: 3; involved: 1, 4)

WP 2	Scalable Distributed Quantum Architectures				Start Month: 1	End Month: 36
Contribution of project partners						
Partner number		1. SNU	2. Yonsei	3. Fraunhofer	4. Naples	
Total effort per partner (person*months)		120	12	8	18	
Aim of the WP						
To develop theoretical protocols for "Virtual Large-Scale" quantum computing by implementing Multi-Chip Ensembles, enabling scalable QML on fragmented NISQ hardware via parallel model aggregation. This WP directly addresses Objective 1 (Scalability) .						
Tasks						
T2.1	Multi-Chip Protocol Definition (M1–M12: SNU; Naples, Fraunhofer)					
	Mathematical formulation of ensemble learning strategies for distributed QPUs; development of aggregation logic (voting/averaging) to combine outputs from independent quantum models; definition of inter-chip communication protocols.					
T2.2	Algorithm Development (M7–M18; SNU; Naples, Fraunhofer)					
	Design of scalable QML architectures (Quantum Transformers, HQTCN) optimized for ensemble deployment; development of Multi-Modal Data Fusion techniques for heterogeneous data (e.g., sMRI on Chip A, fMRI on Chip B).					
T2.3	Simulation & Proof-of-Concept (M13–M24; SNU; Naples, Fraunhofer)					
	Implementation of Multi-Chip protocols in simulation environments; validation of scalability claims on >20 qubit distributed systems; verification of barren plateau mitigation through restricted global entanglement.					
T2.4	Optimization & Refinement (M19–M30; SNU; Fraunhofer)					
	Optimization of qubit-per-chip requirements; refinement based on feedback from WP4 (HEP validation) and WP5 (reliability testing); performance tuning for real-world data characteristics.					
T2.5	Library Integration (M25–M36; SNU; Naples, Fraunhofer)					
	Integration of validated Multi-Chip Ensemble protocols into the unified "Physics-Aware QML" software library; documentation and API standardization.					
Deliverable		Month of delivery	Title of deliverable			
D2.1		M12	Report on Multi-Chip Protocols			
D2.2		M24	Software Module for Distributed Circuit Aggregation			

WP 3		Physics-Informed Optimization & Generative Models			Start month: 1	End month: 36
Contribution of project partners						
Partner number		1. SNU	2. Yonsei	3. Fraunhofer	4. Naples	
Total effort per partner (person*months)		72	4	6	48	
Aim of the WP To solve the "Trainability" and "Noise" bottlenecks by developing the Quantum Forward-Forward algorithm (QFF) driven by Hybrid Quantum Genetic Algorithms (HQGA), Quantum State Space Models (Q-SSM) for temporal expressibility, and Fuzzy-controlled Quantum Diffusion models. This WP addresses Objective 2 (Trainability) , Objective 3 (Temporal Expressibility) , and Objective 4 (Reliability) .						
Tasks						
T3.1		QFF-HQGA Protocol Definition (M1–M12; Naples; SNU) Development of Quantum Forward-Forward (QFF) algorithm with local "goodness" objectives; integration with Hybrid Quantum Genetic Algorithms (HQGA) for gradient-free optimization; benchmarking convergence against Backpropagation in Barren Plateau landscapes.				
T3.2		Q-SSM Architecture Design (M7–M18; SNU; Naples) Development of Quantum State Space Models (Q-SSM) integrating variational quantum circuits with classical LSTM-style gating mechanisms; implementation of three-branch quantum superposition for measurement-based interference effects; design of chunked recurrence processing for sequential data.				
T3.3		Fuzzy Quantum Diffusion Development (M13–M24; Naples; SNU) Design of Fuzzy Logic Controllers (FLC) to model NISQ noise as uncertainty; integration of FLC with Positive Operator-Valued Measures (POVMs) for continuous "unsharp" measurements; development of Hardware-Attention U-Net for reverse diffusion process.				
T3.4		Simulation Validation (M13–M30; Naples; SNU, Fraunhofer) Full implementation of QFF-HQGA, Q-SSM, and Fuzzy-Quantum Diffusion models; simulation-based validation of trainability improvements and noise resilience; comparison with classical baselines.				
T3.5		Library Integration (M25–M36; Naples; SNU, Fraunhofer) Integration of all optimization and generative model algorithms into the unified "Physics-Aware QML" library; API development and documentation.				
Deliverable		Month of delivery	Title of deliverable			
D3.1		M18	Algorithm Definition: QFF-HQGA, Q-SSM & Fuzzy-Diffusion			
D3.2		M30	Benchmarking Report: Trainability, Temporal Expressibility & Noise Resilience			

WP 4	High-Complexity Domain Validation (HEP & Neuro)			Start month: 1	End month: 36
Contribution of project partners					
Partner number		1. SNU	2. Yonsei	3. Fraunhofer	4. Naples
Total effort per partner (person*months)		84	48	8	12
Aim of the WP					
To validate the foundational breakthroughs (WP2/WP3) using massive, complex datasets from High Energy Physics (LHC) and Neuroscience (EEG, fMRI), proving quantum advantage in scalability and temporal modeling. This WP addresses Objective 5 (Validation) .					
Tasks					
T4.1	Dataset Curation (M1–M12; Yonsei; SNU, Fraunhofer) Preparation of LHC data including CMS calorimeter images, high-level physics features, and detector waveform signals (Yonsei); curation of EEG/fMRI time-series datasets (SNU); implementation of quantum feature maps for high-dimensional data (Fraunhofer).				
T4.2	Quantum Feature Mapping (M7–M18; Yonsei; SNU, Naples, Fraunhofer) Development of quantum encoding schemes for HEP and Neuroscience data; optimization of feature map architectures for domain-specific characteristics; implementation of dimension reduction strategies compatible with Multi-Chip protocols.				
T4.3	HEP Validation (Multi-Chip + LHC) (M13–M30; Yonsei; SNU, Fraunhofer) Application of Multi-Chip Ensembles (from WP2) to LHC data classification using Quantum Vision Transformers; implementation of Q-ABCDiCo for background estimation; TCN-VQC for waveform denoising; benchmarking accuracy and generalization vs. single-chip and classical SOTA baselines.				
T4.4	Neuro Validation (Q-SSM + EEG/fMRI) (M13–M30; SNU; Naples, Fraunhofer) Application of Q-SSM (from WP3) to EEG/fMRI spatio-temporal analysis; implementation of QFF-HQGA driven QAOA/VQE for brain connectivity optimization; evaluation of long-range temporal dependency capture vs. classical transformers.				
T4.5	Final Benchmarking vs SOTA (M25–M36; Yonsei; SNU, Naples, Fraunhofer) Comprehensive benchmarking of all quantum approaches against classical state-of-the-art; statistical analysis of quantum advantage claims; preparation of validation reports.				
Deliverable		Month of delivery	Title of deliverable		
D4.1		M6	Curated "Challenge Datasets" (LHC & EEG/fMRI)		
D4.2		M36	Validation Report: Roadmap to Quantum Advantage in HEP & Neuro		

WP 5	Reliability, Safety & Certification			Start month: 1	End month: 36
Contribution of project partners					
Partner number	1. SNU	2. Yonsei	3. Fraunhofer	4. Naples	
Total effort per partner (person*months)	24	6	18	18	
Aim of the WP					
To define the first "QML Reliability Standard" and benchmark and certify the robustness of the developed QML models against cybersecurity threats, adversarial attacks, and privacy risks. This WP addresses Objective 4 (Reliability) certification aspects.					
Tasks					
T5.1	Requirements Specification (M1–M12; Fraunhofer; All) Definition and development of reliability and safety requirements for QML models; specification of robustness and reliability metrics (e.g., Lipschitz continuity, noise stability, data leakage); identification of cybersecurity vulnerability and inherent defenses relevant to model types developed in WP2 and WP3.				
T5.2	Robustness Enhancement (M7–M18; Fraunhofer; Naples) Development of formal metrics for QML robustness; design of benchmarking protocols for adversarial perturbation testing; definition of privacy budget and data leakage metrics.				
T5.3	Vulnerabilities and Defenses Testing (M13–M30; Fraunhofer; Naples) Application of Evolutionary QML to Network Intrusion Detection; stress-testing the Fuzzy-Quantum Diffusion models with gradient-based adversarial attacks; adversarial certification for recurrent (Q-SSM) and attention-based models.				
T5.4	Certification Protocol Development (M25–M36; Fraunhofer) Synthesis of results into a formal certification protocol for industrial QML applications; development of certification guidelines for safety-critical deployment; documentation of reliability standards.				
Deliverable		Month of delivery	Title of deliverable		
D5.1		M18	Robustness Report: Metrics & Initial Assessment		
D5.2		M36	QML Reliability Standard & Certification Protocols		

WP 6	Dissemination, Exploitation & Communication			Start month: 1	End month: 36
Contribution of project partners					
Partner number	1. SNU	2. Yonsei	3. Fraunhofer	4. Naples	
Total effort per partner (person*months)	24	8	22	10	
Aim of the WP					
To maximize impact through Open Science publications, release of the "Physics-Aware QML" library, and engagement with the industrial and public sectors.					
Tasks					
T6.1	Communication Planning (M1–M12; Fraunhofer; SNU) Development of project website and communication strategy; establishment of social media presence; creation of project visual identity and materials.				
T6.2	Publications & Conferences (M13–M36; All) Publication of results in Open Access journals (Nature Comms, PRX Quantum, Quantum Machine Intelligence, etc.); management of the GitHub repository and Zenodo data archives; presentations at major conferences (IEEE QCE, CHEP, etc.).				
T6.3	Hackathon & Summer School (M25–M30; Fraunhofer; SNU) Organization of the "Quantum Robustness Hackathon" (M30) challenging participants to attack/defend QML models; summer school on Physics-Aware QML for early-career researchers.				
T6.4	Open-Source Library Release (M31–M36; SNU; Naples, Fraunhofer) Final release of the integrated "Physics-Aware QML" open-source library under Apache 2.0 license; comprehensive documentation and Jupyter notebook tutorials; Zenodo archival with DOI assignment.				
Deliverable		Month of delivery	Title of deliverable		
D6.1		M36	Open Source "Physics-Aware QML" Library Release		
D6.2		M36	Final Plan for Exploitation and Standardization		

Work package overview (total effort per WP and partner in person.months)

Partner	WP1	WP2	WP3	WP4	WP5	WP6	Total
1. SNU	18	120	72	84	24	24	342
2. Yonsei	2	12	4	48	6	8	80
3. Fraunhofer	4	8	6	8	24	22	72
4. Naples	2	18	48	12	18	10	108
Total	26	158	130	152	72	64	602

3.3 Management structure, milestones, risk assessment

Organizational Structure and Decision-Making

Project Coordinator (PC): Prof. Jiook Cha (SNU) is the single point of contact for the QuantERA Secretariat. The PC is responsible for overall progress monitoring, submission of deliverables, and conflict resolution.

General Assembly (GA): The ultimate decision-making body, composed of one representative from each partner. The GA meets bi-annually to review progress against Milestones. Decisions are made by consensus; if consensus fails, a majority vote (one vote per partner) applies.

Scientific Steering Committee (SSC): Composed of the Work Package Leaders (WP1/WP2: SNU; WP3: Naples; WP4: Yonsei; WP5/WP6: Fraunhofer). The SSC meets quarterly to ensure technical alignment across WPs.

Innovation & Ethics Board: Led by Fraunhofer IKS, this board oversees IPR management, Open Science compliance, and ethical handling of sensitive data (Neuroscience EEG/fMRI, Cybersecurity network logs).

Appropriateness of Structure

This structure is appropriate because it is **lean** (minimizing administrative overhead for a 4-partner consortium) yet **robust** (ensuring all scientific disciplines—Physics, Computer Science, Engineering—have equal representation in the GA). The geographic distribution is balanced with clear communication channels to bridge the Asia-Europe collaboration.

List of Milestones

Milestone	Month	WP Involved	Title
MS1	M12	WP2, WP3, WP4, WP5	Theoretical Foundations Established
MS2	M24	WP2, WP3, WP4	Algorithmic Validation (Simulation)
MS3	M30	WP4, WP5	Cross-Domain Integration
MS4	M36	WP2, WP3, WP5, WP6	Final Deliverables

Success Criteria

- **MS1:** Multi-Chip Ensemble protocol specifications complete; QFF-HQGA and Q-SSM architectures defined; Fuzzy Logic Controllers specified; reliability metrics defined; Domain datasets (LHC, EEG/fMRI) preprocessed and feature-mapped
- **MS2:** QFF-HQGA converges on >10-layer circuits where backpropagation fails; Q-SSM achieves $\geq 1.5\times$ classical memory capacity; Fuzzy-Quantum Diffusion achieves FID $\leq 1.5\times$ classical diffusion baseline
- **MS3:** Multi-Chip Ensemble reconstructs LHC particle correlations across 2+ parallel chips; Q-SSM demonstrates superior long-range temporal capture on EEG/fMRI; QML models pass robustness certification against adversarial attacks
- **MS4:** Open-source "Physics-Aware QML" library released (Apache 2.0); Final benchmarks published demonstrating advantage in scalability (O1), trainability (O2), temporal expressibility (O3), and reliability (O4: Lipschitz $L < 2.0$ certification); QUARK Certification Standard documented

Critical Risks for Implementation

Description of Risk	WPs	Proposed Risk-Mitigation Measures
Scientific: Barren Plateaus persist despite QFF-HQGA methods, preventing convergence in deep circuits.	WP3, WP4	Mitigation: We do not rely on a single optimization strategy. The HQGA component provides gradient-free global search via entangled crossover and quantum elitism. If QFF proves unstable, HQGA serves as a robust fallback that navigates the optimization landscape without gradient dependence. Additionally, the local "goodness" objectives of QFF inherently reduce gradient variance.
Technical: Ensemble Aggregation fails to achieve accuracy targets due to weak individual learners on small chips.	WP2, WP4	Mitigation: We will implement advanced "Weighted Voting" mechanisms where each chip's output weight is determined by its reliability score. The Quantum Transformer ansatz is specifically designed to maximize expressibility even on small chips. Selective Entanglement between chips processing globally dependent features will enhance collective expressivity.
Technical: Q-SSM fails to capture long-range temporal correlations in EEG/fMRI data, underperforming classical baselines.	WP3, WP4	Mitigation: The three-branch quantum superposition ($\alpha, \beta, \gamma \in \mathbb{C}$) architecture provides interference-like effects while LSTM-style gating handles sequential memory. If temporal modeling is insufficient, we will enhance the bidirectional processing layer and increase chunk overlap. Alternative QSVT-enhanced attention mechanisms provide a fallback architecture.
Technical: Fuzzy-Quantum Diffusion fails to achieve target generative fidelity under NISQ noise conditions.	WP3, WP4	Mitigation: The Fuzzy Logic Controllers using POVMs (Positive Operator-Valued Measures) are specifically designed to model noise as continuous "degrees of uncertainty" rather than binary errors. If the Hardware-Attention U-Net underperforms, we can pivot to Quantum GANs enhanced with the same Fuzzy Controllers to stabilize adversarial training dynamics.
Operational: Delays in accessing sensitive domain data (LHC physics data, EEG/fMRI clinical data, network security logs).	WP4, WP5	Mitigation: All partners have established direct access: Prof. Yoo is a CMS Collaboration member with CERN LHC access; SNU has IRB-approved neuroimaging protocols. For contingency, open-source proxy datasets (CERN Open Data Portal, PhysioNet EEG archives, KDD Cup 99) are identified to proceed with algorithmic development without delay.
Management: IPR disputes regarding industrial exploitation of jointly-developed algorithms.	WP1, WP6	Mitigation: A comprehensive Consortium Agreement (CA) based on the DESCA model will be signed before project start, clearly defining background/foreground IP ownership and licensing terms. The "Open Source by Default" strategy (Apache 2.0) for the core library minimizes commercial disputes while preserving academic recognition.

3.4 Consortium as a whole (max. 1 page)

Matching Objectives and Expertise

The consortium is designed as a **trans-continental feedback loop** bridging Foundational Quantum Theory (O1–O4) and Rigorous Domain Validation (O5). No single partner possesses the full spectrum of skills required to deliver the "Physics-Aware QML Stack," necessitating this specific four-partner collaboration:

Role	Partner	Objective Addressed	Key Contribution
Architects	SNU (Korea)	O1: Scalability	Multi-Chip Ensemble protocols, Q-SSM, Quantum Transformers
Optimizers	Naples (Italy)	O2: Trainability, O4: Reliability	QFF-HQGA, Fuzzy Quantum Diffusion, POVMs
Validators	Yonsei (Korea)	O5: Domain Validation	LHC/CMS data, Q-ABCDiCo, TCN-VQC, Quantum ViT
Certifiers	Fraunhofer IKS (Germany)	O4: Reliability Certification	Robust QML, Lipschitz bounds, adversarial testing, benchmarking

Complementarity and Contribution

The partners complement each other through a **"Scale-Logic-Trust"** triangle with domain grounding:

- **SNU (Scale)** provides the *structure* of quantum models. They contribute **342 PM** (57% of total effort) across WP1/WP2 leadership. Without SNU, the consortium would lack the scalable architectures (Multi-Chip Ensembles with Selective Entanglement, variance reduction $\propto 1/k$) to run on fragmented NISQ hardware.
- **Naples (Logic)** provides the *training engine*. They contribute **108 PM** leading WP3. Without Naples, SNU's deep architectures would be untrainable (Barren Plateaus) or fragile to noise. Their QFF-HQGA (entangled crossover, quantum elitism) and Fuzzy Logic Controllers using POVMs provide the "soft" resilience layer enabling hardware-native generative modeling.
- **Fraunhofer IKS (Trust)** provides the *certification standard*. They contribute **72 PM** leading WP5/WP6. Without Fraunhofer, results would remain academic proof-of-concept. A robustness analysis transforms outputs into certified software assets (Lipschitz $L < 2.0$ guarantees) ready for safety-critical deployment.
- **Yonsei (Physical Reality)** grounds the project in *domain science*. They contribute **80 PM** leading WP4. Their CMS Collaboration membership provides direct access to LHC petabyte-scale data with intrinsic non-local correlations—the ideal testbed for proving that Multi-Chip Ensembles preserve quantum expressivity. Without Yonsei, algorithms would lack the "Big Science" validation complexity that distinguishes foundational claims from toy demonstrations.

Resource Adequacy: The total **602 Person-Months** distributed across partners ensures each has adequate capacity: SNU's large team (19 researchers) handles the dual burden of coordination (WP1) and core architecture development (WP2); Naples' focused team (4 researchers) provides specialized optimization expertise; Fraunhofer's industrial researchers (2 senior staff) bring application-focused certification experience; Yonsei's physics team (4 researchers) manages the demanding HEP data pipeline.

Industrial/Commercial Involvement

While the consortium focuses on foundational QPR research, **Fraunhofer IKS** serves as the bridge to industry. As a leading German Research and Technology Organisation (RTO), Fraunhofer's involvement ensures the exploitation pathway is built-in from project inception:

- **Munich Quantum Valley Integration:** Fraunhofer IKS coordinates the European QC Benchmarking Coordination Committee and leads the Munich Quantum Valley consortium on quantum computing applications, providing direct channels to industrial stakeholders in automotive, aerospace, and medical technology sectors.
- **Robust QML as Industry Standard:** The "QML Reliability Standard" (WP5, D5.2) directly aligns with Fraunhofer's mandate. By previous research to certify the consortium's algorithms with formal metrics (gradient variance, entangling capability, Lipschitz bounds), Fraunhofer validates commercial viability of "Trustworthy QML" services—enabling the consortium to offer **certified algorithms** rather than experimental demonstrations.
- **Exploitation Consistency:** This industrial involvement is consistent with the proposed exploitation measures (Section 2): Fraunhofer can immediately translate certification protocols into consulting services for companies requiring safety guarantees before QML deployment. The "Open Source by Default" strategy (Apache 2.0) for the core library maximizes academic impact while the certification layer creates commercial value.

3.5 Description of the consortium

(max. 1 page per partner)

Describe expertise and role in the project for each partner (templates provided). The information provided here will be used to judge the operational capacity. Use the following templates for the coordinator, the other partners requesting funding, and partners not requesting funding if any. If the project relies on input to be provided by a third party, append a letter of commitment at the end of the proposal.

Jiook Cha Project Coordinator	Seoul National University / Department of Psychology, Department of Brain and Cognitive Sciences, Interdisciplinary Program in Artificial Intelligence
Expertise: Seoul National University (SNU) is South Korea's premier research university. Prof. Jiook Cha's lab specializes in Quantum Machine Learning (QML) architectures and their application to Computational Neuroscience . The group possesses unique expertise in:	

1. **Multi-Chip Ensembles:** Developing scalable approaches to train QML models across multiple chips, applied to image data, EEG/fMRI, and Reinforcement Learning.
2. **Quantum Time-Series Analysis:** Pioneering quantum algorithms for temporal data, including Quantum Transformers for fMRI analysis.
3. **QML for Neuroscience:** Bridging quantum computing with brain science through specialized neural network architectures.

Principal Investigator: Prof. Jiook Cha (Male)

Prof. Cha is a leading expert in applied QML architectures. His research focuses on creating scalable quantum algorithms for high-dimensional biological data.

Relevant Achievements:

- Developing **multi-chip ensembles** approaches and applying them to image data, EEG data, and reinforcement learning environments.
- Developing **quantum time-series transformers** and applying them to fMRI analysis.
- Developing **hybrid quantum temporal convolutional neural network (HQTCN)** algorithms and applying them to EEG analysis.

Role in project: Coordinator & QML Architecture Lead.

SNU leads WP1 (Management) and drives WP2 (Scalable Architectures) by implementing the Multi-Chip Ensemble protocols. Prof. Cha also leads the theoretical development of the Neuro-Generative validation in WP4, utilizing his expertise in Quantum Transformers and HQTCNs to process EEG/fMRI data.

Use this template if this partner is requesting funding

Hwidong Yoo	Yonsei University / Department of Physics
Expertise: Yonsei University is a leading private research university in Korea with a strong legacy in High Energy Physics (HEP). Prof. Hwidong Yoo is a key member of the CMS Collaboration at CERN, providing the consortium with direct access to massive-scale "Big Science" data. 1. High Energy Physics: Deep expertise in LHC particle jet analysis, tracking, and event classification. 2. Big Data Analysis: Infrastructure for processing petabyte-scale datasets from particle collisions. 3. Quantum for HEP: Experience in benchmarking quantum algorithms against classical Deep Learning (CNNs/RNNs) for particle physics tasks. Principal Investigator: Prof. Hwidong Yoo (Male) Prof. Yoo is an expert in experimental high-energy physics and the application of advanced computing to particle detection. He leads the validation efforts, bridging the gap between abstract quantum algorithms and physical reality. Relevant Achievements: • Leadership roles in the CMS Collaboration (CERN) for particle data analysis. • Development of classical baselines for CMS image, feature, and waveform analysis tasks. • Pioneering work in applying Quantum Vision Transformers, Quantum Neural Networks, and hybrid QML models to CMS classification, background estimation, and waveform denoising.	
Role in project: Domain Validation Lead (HEP). Yonsei leads WP4 (High-Complexity Domain Validation). They are responsible for curating the LHC datasets, implementing the quantum feature maps for high-dimensional particle data, and rigorously benchmarking the Scalable QML protocols (WP2) against classical SOTA methods.	

Jeanette Miriam Lorenz	Fraunhofer / Fraunhofer Institute for Cognitive Systems IKS
Expertise: Fraunhofer IKS is Germany's leading applied research institute for Safe and Reliable Software Engineering and Artificial Intelligence. PD Dr. habil. Jeanette Miriam Lorenz's group specializes in Robust and Reliable Quantum Computing for Applications, focusing on a broad range of topics encapsulating QML, quantum optimization, quantum simulations and physics-inspired methods.	

1. **Application-driven QML:** Broad expertise in key QML performance aspects, including data-efficiency, impact of feature maps' design, noise stability and generalization.
2. **Quantum Robustness and Safety:** Extensive hands-on experience in QNN certification and verification against noise impacts and cyberattacks, which are critical for the project's cybersecurity validation and "Quality of Service" standards
3. **Benchmarking:** Fraunhofer IKS is one of the European's leaders in application-driven benchmarking of QCs, coordinates the European QC Benchmarking Coordination Committee, and engages in co-developing the QUARK framework for .

Principal Investigator: PD Dr. habil. Jeanette Miriam Lorenz (Female)

PD Dr. Lorenz heads the department 'Quantum Computing' at Fraunhofer IKS and is associate professor at LMU Munich. She also leads the consortium focussing on applications of quantum computing in the Munich Quantum Valley. The mission of her research is to enable the use of quantum computing in safety-critical areas such as aerospace, mobility, logistics and medtech.

Relevant Achievements:

- Establishes the guidelines to perform application-centric benchmarking of QC within Europe, e.g. via the EQCBC and the QUARK-framework
- Extensive work on security and safety of QML, e.g. for the BSI (Federal Ministry of Information Security)
- Bridging theoretical work with practical aspects of 'getting quantum computing to work'

Role in project: Reliability & Certification Lead

Fraunhofer leads **WP5 (Reliability, Cybersecurity & Certification)** and **WP6 (Dissemination)**.

They define the **Quantum Reliability Standards**, and conduct the rigorous **Cybersecurity stress-tests** (Adversarial Attacks) to certify the consortium's algorithms.

Giovanni Acampora	University of Naples Federico II / Department of Physics "Ettore Pancini"
Expertise:	

The University of Naples Federico II is one of the oldest universities in the world. Prof. Giovanni Acampora's group is globally recognized for **Computational Intelligence** and **Fuzzy Logic**. They bring the critical "classical intelligence" layer to the quantum stack.

1. **Fuzzy Logic Systems:** World-class expertise in Type-2 Fuzzy Sets and Fuzzy Logic Controllers, essential for the project's noise-resilience goals.
2. **Evolutionary Computation:** Deep experience in Memetic Algorithms and Bio-inspired optimization to solve non-convex problems.
3. **Quantum Computational Intelligence:** Pioneering the fusion of Fuzzy Logic with Quantum Computing (Quantum Fuzzy Logic).

Principal Investigator: Prof. Giovanni Acampora (Male)

Prof. Acampora is a Senior Member of IEEE and a pioneer in Quantum Computational Intelligence. His work focuses on hybridizing soft computing with quantum mechanics to solve trainability issues.

Relevant Achievements:

- Definition of **Quantum Fuzzy Logic** frameworks for uncertainty management.
- Development of **Evolutionary strategies** for optimizing Variational Quantum Circuits.
- Application of **Memetic Algorithms** to prevent Barren Plateaus in QML.

Role in project: Optimization & Intelligence Lead

Naples leads **WP3 (Physics-Informed Optimization)**. They provide the **Evolutionary/Memetic algorithms** to drive the QFF training and design the **Fuzzy Logic Controllers** for the Robust Quantum Diffusion models. They also support WP5 in adversarial evolutionary testing.

3.6 Consortium agreement principles (partner's rights and duties, IPR management, data management, open access, GDPR)

(max. ½ page)

Consortium Agreement & Decision Making

The consortium will sign a Consortium Agreement (CA) based on the DESCA (Horizon Europe) model prior to the project start. The CA will define the rights and duties of partners, specifying that SNU (Coordinator) manages the administrative interface, while the General Assembly (one vote per partner) decides on major scientific pivots.

Intellectual Property Rights (IPR) Management

- **Background IP:** Partners retain ownership of their pre-existing know-how. Specifically, **Fraunhofer IKS** retains all rights to the core **QUARK framework**, and **Univ. Naples** retains rights to their pre-existing Fuzzy Logic libraries.
- **Foreground IP:** The new "Physics-Aware QML" algorithms (e.g., Multi-Chip Ensemble protocols, Fuzzy-Diffusion architectures) developed jointly will be jointly owned. We adopt an "**Open Source by Default**" strategy for software outputs, releasing the unified library under an **Apache 2.0 license** to maximize adoption.
- **Commercial Exploitation:** Fraunhofer IKS leads the exploitation strategy. If a patentable invention arises (e.g., a specific industrial control system based on Robust QML), the CA outlines fair access rights and royalty-sharing mechanisms for the inventors.

Data Management & Open Access

- **FAIR Data:** We adhere to the **FAIR principles** (Findable, Accessible, Interoperable, Reusable). All non-sensitive "Challenge Datasets" (LHC jets, synthetic EEG) will be deposited in **Zenodo** with DOIs.
- **Open Access:** The consortium commits to **100% Open Access** for peer-reviewed publications (Green or Gold), ensuring immediate availability on **arXiv**.

GDPR & Ethical Compliance

- **Neuroscience Data:** The processing of EEG/fMRI data (WP4) involves human subjects. **SNU** will ensure all data is **pseudonymized** at the source before sharing within the consortium, in strict compliance with **GDPR** and Korean Personal Information Protection Act (PIPA).
- **Cybersecurity Data:** Network logs used in WP5 will be screened to ensure no personally identifiable information (PII) or critical infrastructure vulnerabilities are exposed.

3.7 Significant facilities and large equipment available to the consortium to perform the project

(max. ½ page)

The consortium brings together world-class infrastructure that enables "Big Science" validation without the need for external hardware procurement.

1. High-Performance Computing & Simulation (SNU & Naples)

- **SNU:** Access to extensive GPU clusters required to simulate the **Multi-Chip Ensemble** protocols. Since we are simulating distributed quantum learning (validating >20 qubits across multiple virtual chips), high-memory classical simulation resources are critical.
- **Univ. Naples:** Dedicated computational clusters optimized for **Evolutionary Algorithms**, which require massive parallel processing capabilities to evolve the populations for the Memetic QFF optimization tasks.

2. "Big Science" Data Infrastructure (Yonsei)

- **Yonsei (CMS Collaboration):** Through Prof. Yoo's leadership in the CMS Collaboration, the consortium has authorized access to the **CERN LHC computing**

grid and petabyte-scale storage systems. This provides the massive **High Energy Physics datasets** (particle jets) required for WP4, which are otherwise inaccessible to standard research groups.

3. Reliability Benchmarking & Quantum Hardware Access (Fraunhofer IKS)

- **The QUARK Framework:** Fraunhofer provides the proprietary **QUARK benchmarking suite**, a significant software facility that automates the assessment of reliability metrics (Lipschitz continuity, robustness). This pre-existing infrastructure accelerates WP5 significantly.
- **Munich Quantum Valley (MQV):** As a key member of MQV, Fraunhofer IKS provides access to a diverse ecosystem of quantum hardware backends (Superconducting, Ion Trap) for validating the **Multi-Chip** protocols on physical NISQ devices where applicable.

3.8 Link with ongoing projects

(max. ½ page)

For each partner indicate (if applicable) the ongoing projects linked to the proposal topic, and their funding sources.

3.9 Financial plan

(max. 1 page)

The resources to be committed for each project partner have to be described in the Electronic Submission System by the coordinator. These resources include: Personnel, Consumables, Equipment, Travel, Subcontracting, Provisions, Licensing fees, other. Justify them here. Both the justification and the information in the system will be communicated to the Evaluation Panel.

The total requested budget for the consortium is **€1,065,945.34**, distributed to ensure the successful execution of the "Physics-Aware QML" work program.

1. Personnel Costs (€569,211.76)

This is the largest budget category, reflecting the labor-intensive nature of foundational algorithmic research.

- **Partner 1 (SNU):** Requesting **€163,905.88** to support full-time PhD students dedicated to developing the **Multi-Chip Ensemble** protocols (WP2) and **Quantum Transformers** (WP4).
- **Partner 2 (Yonsei):** Requesting **€84,705.88** for three graduate students to handle the massive data preprocessing and benchmarking of **LHC particle jets** (WP4).
- **Partner 3 (Naples):** Requesting **€56,000** for a Post-doc researcher specializing in **Evolutionary Algorithms** and **Fuzzy Logic** to lead the optimization tasks in WP3.

- **Partner 4 (Fraunhofer IKS):** Requesting **€264,600** for a senior researcher/Post-doc with industrial experience to drive the **QUARK certification** and **Reliability Standards** (WP5).

2. Equipment (€71,098.04)

Essential hardware is required for simulating quantum circuits and training hybrid models.

- **Partner 1 (SNU) & Partner 3 (Naples):** Requesting **€16,833.33** and **€12,500** respectively for **high-performance GPU workstations**. These are critical for simulating the **Quantum Forward-Forward (QFF)** algorithm, which requires intensive classical matrix operations during the "local goodness" updates (WP3).
- **Partner 2 (Yonsei):** Requesting **€11,764.71** for servers to store and process the petabyte-scale **LHC datasets** locally before quantum feature mapping (WP4).
- **Partner 4 (Fraunhofer IKS):** Requesting **€30,000** specifically for **"Access to Quantum Computers."** This budget is allocated to purchase compute time on cloud-based NISQ backends (e.g., IBM Quantum, IonQ) to validate the **Robustness metrics** on real hardware (WP5).

3. Travel (€162,122.36)

Travel costs are strictly necessary for ensuring cohesion between the Asian and European partners.

- **Partner 1 (SNU) & Partner 2 (Yonsei):** Requesting **~€70,000 each**. This higher amount is justified by the long-distance travel from South Korea to Europe for **bi-annual consortium meetings** and "Methodology Swap" workshops (WP6). It also covers travel for PhD students to attend major conferences (e.g., QIP, IEEE CEC) to disseminate results.
- **Partner 3 (Naples) & Partner 4 (Fraunhofer IKS):** Requesting **€7,000** and **€15,000** respectively for intra-European travel to consortium meetings and the final "Quantum Robustness Hackathon" in Munich.

4. Other Goods, Services & Consumables (€56,104.71)

- **Publication Fees:** A significant portion (approx. **€3,500** per partner) is allocated for **Gold Open Access** publication fees to ensure immediate availability of our foundational results, complying with Open Science mandates.
- **Partner 2 (Yonsei):** Requesting **€11,764.71** for consumables related to data storage media and minor research supplies for the HEP validation experiments.

5. Overheads (€207,408.47)

Calculated according to the specific national rules of each funding agency (NRF Korea, MUR Italy, DFG Germany).

Summary of Resource Commitment

The allocation ensures that SNU and Yonsei have the personnel and travel resources to bridge the geographical gap and drive the heavy theoretical/experimental load, while Fraunhofer IKS is adequately resourced to access real quantum hardware for industrial-grade certification. Univ. Naples is efficiently budgeted to provide the specialized optimization logic.

4. Ethical issues (max. ½ page)

Describe any foreseeable ethical issue that may arise during the course of the research project. Describe all mitigation strategies employed to reduce ethical risk, and justify the research methodology with respect to ethical issues.

Foreseeable Ethical Issues

This project involves two sensitive domains that require rigorous ethical oversight:

1. **Human Biological Data (Neuroscience):** The processing of EEG and fMRI data (WP4) raises concerns regarding participant privacy, data protection (GDPR), and potential re-identification risks.
2. **Dual-Use Potential (Cybersecurity):** The development of adversarial attack protocols (WP5) and robust AI models involves "dual-use" technologies that, while designed for defense, could theoretically be misused to bypass existing security systems.

Mitigation Strategies

We implement a "Safety-by-Design" approach to mitigate these risks:

- **Privacy-Enhancing Technologies (PETs):** For Neuroscience data, we strictly adhere to **GDPR** and the **Declaration of Helsinki**. All data will be pseudonymized at the source (SNU/Naples) before being processed by the quantum algorithms. Furthermore, a core scientific objective of this project is the development of **Fuzzy-Quantum Diffusion Models** (WP3) specifically to generate high-fidelity *synthetic* data. This methodology serves as a direct ethical mitigation, allowing us to share and process realistic brain data without exposing sensitive patient information.
- **Responsible Cybersecurity Research:** To mitigate dual-use risks, we strictly follow **Responsible Disclosure** guidelines. Detailed specifications of the adversarial attack vectors will be shared only within the consortium via encrypted channels ("As Closed As Necessary"). Public dissemination will focus on the *defense mechanisms* (Robustness Certification) rather than the attack tools.

- **Oversight:** An **Innovation & Ethics Board** (led by Fraunhofer IKS) will monitor compliance throughout the project lifecycle, ensuring that all AI models are tested for bias (e.g., sex-specific variances in EEG) before validation.

Justification of Methodology

The use of high-dimensional human data is scientifically necessary to validate the Temporal Expressibility of our Quantum State Space Models (Q-SSM). Synthetic or toy datasets lack the complex, long-range correlations required to prove quantum advantage. Similarly, simulating adversarial attacks is the only mathematically rigorous way to certify the Reliability of QML models using the QUARK framework, a necessary step to ensure future quantum technologies are safe for society.

5. References (max. 30 references)

Provide references of articles and publicly available documents directly supporting the proposal.